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PyFin-sentiment: Towards a machine-learning-based model for deriving sentiment from financial tweets

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ABSTRACT

Responding to the poor performance of generic automated sentiment analysis solutions on domain-specific texts, we collect a dataset of 10,000 tweets discussing the topics of finance and investing. We manually assign each tweet its market sentiment, i.e., the investor's anticipation of a stock's future return. Using this data, we show that all existing sentiment models trained on adjacent domains struggle with accurate market sentiment analysis due to the task's specialized vocabulary. Consequently, we design, train, and deploy our own sentiment model. It outperforms all previous models (VADER, NTUSD-Fin, FinBERT, TwitterRoBERTa) when evaluated on Twitter posts. On posts from a different platform, our model performs on par with BERT-based large language models. We achieve this result at a fraction of the training and inference costs due to the model's simple design. We publish the artifact as a python library to facilitate its use by future researchers and practitioners.

1. Introduction

The advent of social networking sites presents a unique opportunity to tap into an enormous stream of data that users share with the world. Among others, sentiment analysis (also known as emotion artificial intelligence or opinion mining) which implies systematic identification and quantification of affective states (emotions) from text, has been widely used by scholars and practitioners to derive actionable insights across domains, e.g., political communication (Luo & Mu, 2022), tourism industry (Obembe, Kolade, Obembe, Owoseni, & Mafimisebi, 2021), or health records (Chintalapudi, Battineni, Di Canio, Sagaro, & Amenta, 2021). While it is possible to employ human annotators for emotion recognition in a text (Luo & Mu, 2022), the feasibility of this approach is limited to small-scale research experiments. Manual processing speed is incomparable to real-time data processing applications, where performance is measured on the millisecond scale. While many automated sentiment analysis solutions are available, most designs are rooted in generic texts and fail when presented with a domain-specific task.

This work focuses on the finance domain and aims to evaluate how well existing models recognize market sentiment, i.e., positive, negative, or neutral investor anticipation about a company's future stock price development. Initially, the investor mood was mainly gauged through volatility-based indicators like the Chicago Board Options Exchange Volatility Index (VIX) and the Put/Call Ratio (PCR) (for review, see Aggarwal, 2019). However, with the rise of behavioral finance, which accounts for human biases in decision-making processes (Hirshleifer, 2015), the field has started recognizing that retail investors' emotions, sentiments, and opinions also carry valuable information. Previous research has shown that social sentiment obtained from microblogging platforms can help forecast stock market volatility (Antweiler & Frank, 2004; Audrino, Sigrist, & Ballinari, 2020), trading volume (Oliveira, Cortez, & Areal, 2017) and even future returns (Ahuja, Rastogi, Choudhuri, & Garg, 2015; Mittal & Goel, 2012; Ren, Wu, & Liu, 2018; Wilksch & Abramova, 2022). All of these use cases can benefit from more accurate automated sentiment analysis models.

Against this background, the goal of this work is to develop a new model that researchers and practitioners can use to mine retail investors' market sentiment from tweets. The model we propose is unique as it is tailored to the domain of finance-related social media posts and can thus cope with the vocabulary used in such texts. This allows our model to outperform existing artifacts in both predictive power and speed. Moreover, we publish our machine-learning-based model artifact as an easy-to-use python library to foster its application in future studies. We thereby fill an important gap in the existing research where the few functioning model artifacts that are publicly available are either dictionary- or deep-learning-based. To achieve this goal, we formulate four research questions (RQ) which our work seeks to answer.

RQ1: How can we design a functional model artifact that can extract an author's sentiment from finance-related social media posts?

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RQ2: How does this model artifact perform compared to existing models from either the domain of finance-related texts or generic social media posts?

RQ3: Can a small, domain-specific model outperform more generic LLMs?

RQ4: How does the performance of models trained on Twitter posts change when applied to StockTwits posts?

The remainder of this paper is structured as follows. In the *Related Work* section, we survey the literature on existing sentiment analysis technologies and challenges and provide an overview of existing model artifacts that are frequently used in research studies. The *Material and Methods* section lays out our process of collecting and labeling a dataset as well as designing experiments to train and benchmark machine learning and deep learning models on this task. We present the results in the *Results* section and highlight their implications in the *Discussion*.

2. Related work

2.1. Automated sentiment analysis technologies

Previous work on automated sentiment analysis can be categorized as using one of three approaches: dictionary-based analysis, machine-learning-based models, and deep learning approaches. While the discipline of deep learning is a subset of the field of machine learning (Goodfellow, Bengio, & Courville, 2016), we distinguish between the two for this work. Deep learning models require at least an order of magnitude more data, specialized computing resources, and design efforts making them more expensive to train and deploy than simpler machine learning models.

2.1.1. Dictionary-based sentiment analysis

Dictionary-based models use lists of words or phrases to which human researchers have assigned sentiment scores. Simple scoring methods might classify single words as positive or negative, for example LIWC (Pennebaker, Francis, & Booth, 2001), Harvard General Inquirer (Stone & Hunt, 1963), or Opinion Observer (Liu, Hu, & Cheng, 2005). Others rate them on more sophisticated numeric scales, like ANEW (Bradley & Lang, 1999), SentiWordNet (Baccianella, Esuli, & Sebastiani, 2010), or VADER (Hutto & Gilbert, 2014). The sentiment for a document is subsequently calculated as an aggregate of all word scores. This methodology makes dictionaries explainable and computationally cheap at inference time but does not come without drawbacks: The dictionary needs to be compiled by humans which is time-consuming and requires decisions regarding scales and scoring which significantly impact the performance of the final model. Additionally, the rigid approach of gathering a list of words can fail if documents contain few or none of the words in the list. This makes it especially hard to apply lexicon-based approaches to social media content which is riddled with typos, slang words, and emojis. Models not designed for this type of content often classify texts as neutral, simply for the lack of matching words. Furthermore, it is questionable whether the sentiment for a document should be determined through a simple aggregation of per-word sentiment. Finally, the typical challenges that automated sentiment analysis entails (see Section 2.2) have to be addressed manually. For example, Hutto & Gilbert (2014) designed VADER by integrating a set of heuristics for handling negation, punctuation, and capitalization as degree modifiers of sentiment. While the authors had to invest a significant amount of work into crafting these heuristics, they make VADER particularly attractive to researchers working with social media content: Al-Shabi (2020) shows that the integrated heuristics make VADER outperform all of its competitors on social media content.

2.1.2. Machine-learning-based sentiment analysis

Unlike dictionaries, machine learning models can be trained on large corpora of labeled data, enabling researchers to leave it to mathematical optimization algorithms to assess whether a word influences sentiment positively or negatively. While creating such training sets still requires significant resources and manual labor, framing sentiment analysis as a machine learning problem allows for direct optimization of the correct target. Most applications of sentiment analysis are not concerned with assigning a single sentiment score per word. Rather, the unit of analysis is either a sentence or a short document. Machine learning techniques can directly optimize the objective of correctly classifying as many documents or sentences as possible. This alleviates the need for heuristics on how to aggregate word-based scores on a sentence or document level. The most commonly applied machine learning models for sentiment analysis are Support Vector Machines (SVM), Naïve Bayes classifiers, tree-based models, and logistic regression (Ravi & Ravi, 2015). SVMs have been shown to achieve an accuracy of around 75% on the binary classification task of assigning finance-specific posts on StockTwits a "bullish" or "bearish" label (Renault, 2020). For the same two-class sentiment polarity classification task of tweets that are not domain-specific, they can score accuracies as high as 83% (Mishev, Gjorgjevikj, Vodenska, Chitkushev, & Trajanov, 2020; Tang et al., 2014). Naïve Bayes, as well as tree-based models, exhibit similar performance characteristics on generic texts (Mishev et al., 2020).

2.1.3. Deep-learning-based sentiment analysis

In other areas in the field of natural language processing (NLP), like question answering or natural text generation, all prevailing models are based on deep learning. Accordingly, researchers started applying deep learning to the task of sentiment analysis, which is usually framed as a text classification problem. Almost all deep learning models in the field of NLP are currently leveraging large language models (LLM) which are fine-tuned to specific tasks. LLMs are sizable neural networks that have been trained on immense amounts of data. By training them on tasks like predicting masked words from a surrounding sentence, such models learn intricate patterns of natural language. Therefore, they can be used for other tasks than the one originally trained on. The representations of text that LLMs like "Bidirectional Encoder Representations from Transformers" (BERT) learn can be used by a single layer in a neural network to be fine-tuned on a wide variety of tasks (Devlin, Chang, Lee, & Toutanova, 2018). For sentiment analysis of financial news headlines, Araci (2019) constructs FinBERT, a version of BERT that has been fine-tuned on several corpora of news headlines. It outperforms all other benchmark models and reaches an accuracy of 86% in classifying the headlines as positive, negative, or neutral. Barbieri, Camacho-Collados, Neves, & Espinosa-Anke (2020) developed TwitterRoBERTa, a version of RoBERTa (Liu et al., 2019) that they fine-tuned on generic Twitter sentiment analysis. RoBERTa is based on BERT but improves on key training parameters that enhance performance. This helps TwitterRoBERTa to outperform the SVM- and LSTM-based benchmarks of analyzing sentiment in tweets. Most recently, a combination of machine learning and deep-learning-based models has been shown to achieve outstanding performance for sentiment analysis of generic social media posts (AlBadani, Shi, & Dong, 2022). However, few of these benchmarks consider the larger latency, training-, and deployment costs of neural networks and LLMs in particular.

2.2. Challenges for automated sentiment analysis

Considering the ambiguous nature of sentiments and opinions, analyzing them entails multiple challenges that need to be addressed. According to Hussein (2018), the most common sentiment analysis challenges are negation handling, domain dependence, spam detection, and ambiguity in the form of abbreviations or sarcasm. Negation handling presents an issue because a few words that might not be close to the sentiment-laden part of a sentence can completely invert its meaning. In combination with domain-specific vocabulary, this can even be hard to spot for human annotators. However, domain-specificity is not only

Table 1
Overview of sentiment analysis models by domain.

	Finance-specific	
SNS-specific	Yes	No
Yes	Sohangir et al. (2018)*† NTUSD-Fin Chen et al. (2018)	SentiStrength Thelwall, Buckley, Paltoglou, Cai, & Kappas (2010), AFINN Nielsen (2011), VADER Hutto & Gilbert (2014), Twitter RoBERTa Barbieri et al. (2020)†
No	Loughran & McDonald (2011), FinBERT Araci (2019)†	Harvard-IV-4 Stone & Hunt (1963), ANEW Bradley & Lang (1999), LIWC Pennebaker et al. (2001), Opinion Observer Liu et al. (2005), SentiWordNet Baccianella et al. (2010)

* Model artifact has not been published.
† Deep-learning-based model.

problematic when it occurs in conjunction with negation. Different vocabulary, idioms, slang, and divergent interpretations of common words between domains can significantly degrade the quality of a sentiment analysis. Ravi & Ravi (2015) provide an overview of work that addresses the challenge of cross-domain sentiment analysis but conclude that it is still an unsolved problem. On datasets obtained from social media platforms, the issue of spam detection needs careful consideration. Many posts on social media are advertisements or were created by automated robots that post similar content multiple times. Not only can such duplicates ruin the quality of a collected data set, but they also dilute the content posted by real humans as spammers try to blend in with them as much as possible. Removing spam is viable through heuristics developed after manual inspection of a data set, for example by using word lists (Yao & Wang, 2020). However, researchers must scrutinize the precision of such methods to not remove too much informative human-created content and accept that they will likely not detect 100% of all spam posts. Arguably the hardest challenge is coping with ambiguity and sarcasm. Using text as a medium of exchange of opinions can make these stylistic devices hard to identify even for humans. Some expressions require intonation or other cues to convey whether they are a sarcastic note or a serious opinion. This makes sentiment analysis a problem on which even humans might not unanimously agree. Consequently, the uncertainty that is present in any labeled training dataset carries forward to any model built on this data.

2.3. Available sentiment analysis models and datasets

Table 1 presents an overview of the most commonly used sentiment analysis models in the literature. It organizes them as being applicable to the domain of finance and/or social networking sites (SNS). The models listed in the table are dictionary-based unless noted otherwise. It is evident that historically, many models have been developed for generic, non-social-media-related texts. In recent years, the literature has shifted towards coping with harder-to-analyze social media posts. However, most of the models are not applicable to texts using domain-specific financial vocabulary. While Loughran & McDonald (2011) developed a dictionary based on corporate filings and Araci (2019) presents a model trained on business news headlines, their model performance might suffer when applied to the colloquial language found on social media. For the intersection of sentiment analysis of finance-related social media posts, Sohangir, Wang, Pomeranets, & Khoshgoftaar (2018) train a convolutional neural network that performs well on their data but is not published as a usable model artifact. This leaves NTUSD-Fin (Chen, Huang, & Chen, 2018), a freely available dictionary.

The scarcity of usable sentiment analysis models for this domain can partially be explained by the lack of datasets on which such models can be trained. The only datasets that are related to this task are SemEval-2017 Task 5 (Cortis et al., 2017) and Fin-SoMe (Chen, Huang, & Chen, 2020). SemEval-2017 Task 5 contains a subtask ("subtask 1") which consists of 2510 labeled messages from StockTwits and Twitter. For each message, three annotators assign each company that is mentioned a sentiment score between −1 and 1. The scores are then consolidated by a fourth expert. While it is designed for aspect-based sentiment analysis, it is sometimes used for simpler polarity classification. The *Fin-SoMe* data set published by Chen et al. (2020) consists of 10,000 social media posts from StockTwits, a social network to discuss stock-based investments. The authors labeled every post with its market sentiment.

We aim to fill this research gap by proposing a sentiment analysis model that is applicable to social media posts discussing the topic of finance and investing. We do this by collecting and labeling a dataset of 10,000 tweets discussing these topics. Subsequently, we design, train, and publish a sentiment model and benchmark it against FinBERT (Araci, 2019), VADER (Hutto & Gilbert, 2014), TwitterRoBERTa (Barbieri et al., 2020), and NTUSD-Fin (Chen et al., 2018).

3. Material and methods

3.1. Data collection

For this work, we collect a dataset of posts from investing discussions on Twitter. To identify these discussions, the platform offers "cashtags", an equivalent of hashtags that start with a "$" followed by a company's ticker symbol. We utilize these tags to query the Twitter application programming interface (API) for posts that discuss investment ideas regarding a company's stock.

To make results comparable to the previous literature we will focus on English posts only. Therefore, we use the S&P500 index as a starting point for selecting ticker symbols to include in the search query. From there, we impose a minimum activity filter on each stock ticker: a ticker is only considered to be actively discussed on Twitter if there are more than 100 tweets per day on average mentioning it. We impose this filter because financial sentiment analysis is only a valuable tool when applied to larger corpora of data. It should not be used when low post volume creates the risk of mistaking the opinion of very few people as the "public" sentiment. By using an activity filter, we ensure that the tweets that are being collected are sampled from active discussions which makes the training data more closely resemble the data that the sentiment models will be applied to at inference time. To conduct the filtering, we collect data on the number of tweets per day for every S&P500 ticker during April of 2022. The distribution of activity per ticker symbol is highly skewed. The top 20 tickers account for 53.7% of all tweets about S&P500 companies. According to the April 2022 data, 56 tickers fulfill the minimum activity constraint and account for 70.9% of tweet volume. Out of these 56, we manually exclude 6 tickers (AME, OGN, TEL, AMP, KEY, STX) because while they represent corporations listed in the S&P500 index, they are mostly used to reference cryptocurrencies on Twitter. The final search query can be found in Appendix A.

Using the final search query, we collect all tweets using the Twitter API's endpoint `/2/tweets/search/all`. We query all tweets posted after April 1, 2021 (00:00:00 UTC) and before May 1, 2022 (00:00:00 UTC). The presented query yields 3,757,384 raw results which are saved and will undergo further filtering and preprocessing. By collecting a little more than one full year's worth of tweets we cover one full business cycle and prevent the collected data from being biased towards a small window of time, for example, earnings season.

3.1.1. Data sampling

For labeling, we randomly sample 10,000 documents. We clean the entire dataset before selecting the subsample to be labeled. This ensures that time invested in labeling is not wasted by handling large amounts of spam posts that could have been removed automatically.

We start by removing all hyperlinks from tweets as they do not constitute natural language. This will be important for subsequent filtering

Table 2
Sample size during data cleaning stages.

Step	*n* tweets after step	Δ
1. data collection	3,757,384	–
2. drop duplicates	3,286,380	−471,004
3. filter number of cashtags & hashtags	2,797,620	−488,760
4. remove spam by ratios	2,774,245	−23,375
5. remove cryptocurrency posts	2,755,824	−18,421

operations which rely on word counts. Next, we remove all duplicates from the dataset. There are two types of duplicates we filter. First, we filter duplicates based on the tweet IDs in case the API returns duplicate results. Second, we remove all tweets that have duplicate texts which are longer than 5 words since a lot of the content on Twitter is generated by bots posting the same tweet multiple times. We choose this threshold because duplicated short tweets can be legitimate messages (for example: "bought $TSLA"). If two tweets longer than five words are duplicated, however, they are most likely a boilerplate message posted by an automated account.

Next, we filter tweets based on the number of hashtags and cashtags. A manual inspection reveals that spam tweets often use many different hashtags or cashtags to appear in as many searches as possible. Therefore, we exclude all tweets containing five or more cashtags or eight or more hashtags. At this point, however, the data that is left still contains numerous spam tweets. Most of them are shorter tweets with relatively many hashtags or cashtags, but not enough to be removed by the previous filter. Hence, we impose another filter based on the ratio of cashtags to words, hashtags to words, and mentions of other users to words. We require each of these ratios to be lower or equal to 0.5 such that a tweet must contain at least as many words as cashtags, hashtags, and mentions.

Finally, the only form of unwanted tweets that still accounts for a significant amount of data is tweets about cryptocurrencies. Similar to Yao & Wang (2020), we define a list of keywords that are frequently used by the cryptocurrency communities on Twitter and require there be less than or equal to two of these keywords in any tweet for it to be included in the final dataset. We allow for two keywords as we want to be conservative in removing data at this stage and stock market investors might also invest in cryptocurrencies. However, most tweets with three or more of these words are irrelevant. The keywords that were generated by iterative manual inspection of the filtering results are *bitcoin, etherium, btc, eth, nft, token, wallet, web3, airdrop, wagmi, solana, opensea, cryptopunks, uniswap, lunar, hodl, binance, coinbase, cryptocom,* and *doge*. Table 2 displays how the filtering stages reduce the sample size *n*.

3.1.2. Data labeling and preprocessing

Following Chen et al. (2020), who point out that market sentiment and general text sentiment need to be treated as two distinct dependent variables, we assign each tweet its market sentiment. To demonstrate the difference between the two, consider the sentence "Nice, I already made a lot of money this morning and just shorted $AAPL, this is gonna be great!". The general sentiment in this document is positive as the author mentions previous successful trades and a great future. However, the author's market sentiment in this sentence is negative. They opened a short position in Apple Inc. (cashtag $AAPL) which only yields a positive return if the stock price of Apple declines. The author, therefore, expects a decline in the market value of Apple shares which we consider a negative market sentiment. We model market sentiment rather than general sentiment because market sentiment information is more valuable for domain-specific analyses. Moreover, existing generic sentiment models like VADER (Hutto & Gilbert, 2014) already perform well for generic sentiment classification which relies on simpler keywords like "great" or "nice" (positive) rather than domain-specific vocabulary like "short" (negative). Using this approach we label each tweet as containing either bullish (positive), bearish (negative), or neutral sentiment according to a codebook which can be found in Table 5 in Appendix B.

Before using the data for training machine learning models, we preprocess it to facilitate the learning of generalizable patterns. We replace all cashtags with the word "TICKER", all mentions of usernames by "@user", all digits by the number "9", all new line characters by spaces, and convert the text to lowercase. Without these steps, machine learning models would be prone to overfitting patterns in the training data. The preprocessing steps encourage the learning of more generalizable patterns, for example, that "TICKER moved +9.9%" refers to a relative price increase, which is more valuable than a model memorizing the pattern "$TSLA moved +4.2%". Combatting overfitting is a major concern as our goal is to build a generalizable model that other researchers can utilize on different datasets.

3.2. Experimental design

3.2.1. Model training

Next, we train multiple machine learning models on the cleaned data. We will compare two machine learning models (a logistic regression and a support vector machine) against three deep learning models (a recurrent neural network and a transformer neural network trained from scratch and a BERT-based classification model). We experiment with both simple and complex models as the simpler models are fast and provide a good performance baseline. Considering that most models in NLP are deep-learning-based, however, we add the two most common architectures for text classification to our experiments and train them from scratch. As a comparison to the BERT-based Fin-BERT and Twitter RoBERTa, we fine-tune our own BERT-based model.

For the machine learning models, the text is split into tokens which are then represented as a matrix of TF-IDF scores which is fed to the models. We utilize the model and vectorizer implementations from *scikit-learn* (Pedregosa et al., 2011) and optimize the most important hyperparameters using *optuna* (Akiba, Sano, Yanase, Ohta, & Koyama, 2019). The hyperparameters we tune are the type of tokenizer (word- or sub-word-based), the n-gram range, the minimum occurrence threshold for each token in the document, and the model's ℓ_2 regularization parameter. Additionally, for the SVM, we tune the used kernel function and kernel degree.

The deep learning models are trained using *PyTorch* (Paszke et al., 2019). For the two neural nets trained from scratch, we stick to subword tokenization (Kudo, 2018) with a vocabulary size of 3000. The general architecture for both models is similar: First, an embedding layer embeds the tokens, which, after a dropout operation, are passed to the recurrent or transformer layer, respectively. The output is processed by one hidden layer before passing it through another dropout operation and then to the output layer which classifies the text. For the recurrent network, we utilize a layer of gated recurrent units (GRU) (Cho, Van Merrienboer, Bahdanau, & Bengio, 2014) and tune their hidden dimensionality, the embedding dimensionality, the token dropout after the input layer, the hidden layer dimensionality, and the dropout before the output layer. Similarly, for the transformer model, we tune the embedding dimensionality, the transformer feed-forward dimensionality, the hidden layer dimensionality, and both dropouts. For the third deep learning model, we use DistilBERT (Sanh, Debut, Chaumond, & Wolf, 2019) which transforms each text into a 768-dimensional vector representation. This vector is then passed through a dropout operation, a hidden layer, another dropout operation, and finally the 3-class output layer. We fine tune the hidden layer's dimensionality as well as the dropout percentage. We use rectified linear unit (ReLU) activations after all hidden layers and train the models using the AdamW optimizer (Loshchilov & Hutter, 2017) with a learning rate of 0.001 and batch size of 64 for a maximum of 50 epochs or until the validation loss plateaus for at least ten epochs.

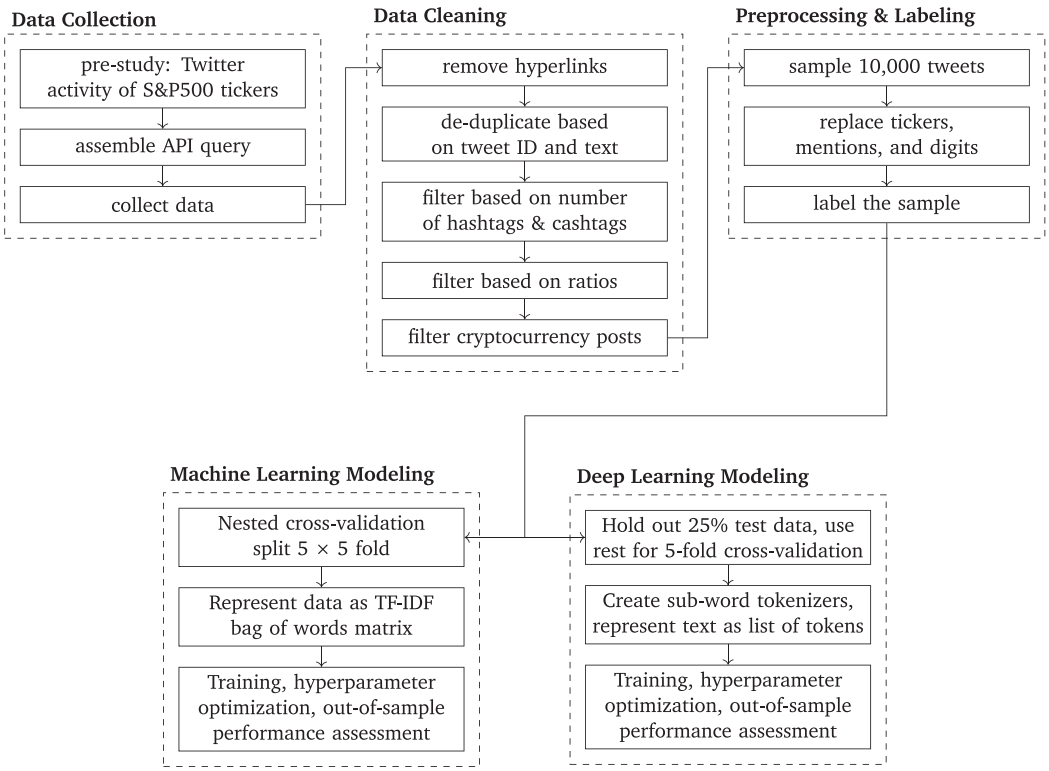

Fig. 1. Flowchart of data collection, preparation, and modeling workflow.

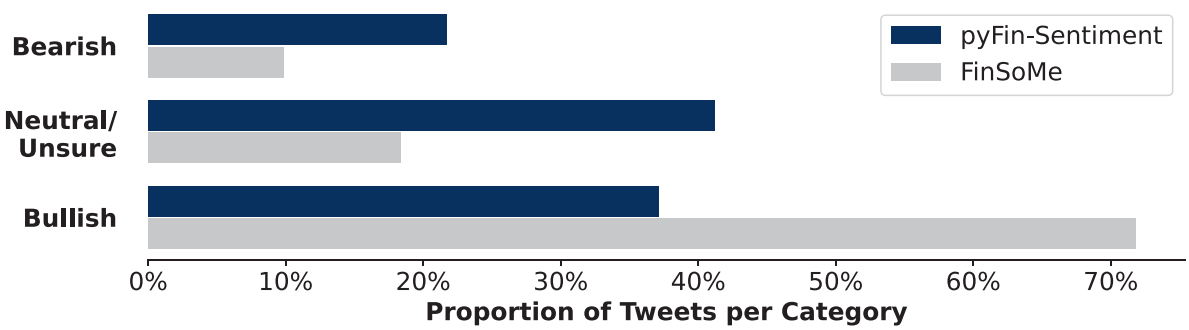

Fig. 2. Class distribution in our dataset vs. Fin-SoMe (Chen et al., 2020).

3.2.2. Model evaluation

Optimizing hyperparameters and obtaining true out-of-sample estimates for a model's performance requires a three-fold data split into a training, validation, and test set. Considering our small dataset size ($n = 10,000$), we apply nested cross-validation (CV) to achieve these goals. We will use an outer 5-fold CV for estimating the models' performance on unseen data. All data that does not belong to this test set split will be used for choosing hyperparameters based on an inner 5-fold CV. All results we report are averages and standard deviations across the five outer test splits. We use these same test splits for benchmarking existing models for which training is not necessary. Due to our limited computing budget, we can not apply the nested cross-validation to the three deep-learning-based models. For them, we use 25% of the data as a hold-out test set and perform normal 5-fold cross-validation on the remaining data for hyperparameter optimization. We compare all models against each other using the Area Under the Receiver Operating Characteristic Curve (ROC AUC) as sentiment class distributions can vary between datasets, in which case the accuracy score can be deceiving. For all model types, we subsequently present the optimal configuration found using the hyperparameter search which explored 100 parameter configurations per split.

The other sentiment analysis models we will benchmark are VADER (Hutto & Gilbert, 2014) and TwitterRoBERTa (Barbieri et al., 2020) from the domain of social media, FinBERT (Araci, 2019) from the domain of finance news, and NTUSD-Fin (Chen et al., 2018) which has been trained on finance-related social media posts from StockTwits. All models will be applied to two relevant datasets: the one we collect in this study, as well as Fin-SoMe (Chen et al., 2020). For the BERT-based FinBERT and TwitterRoBERTa models, we utilize their implementations in the *huggingface transformers* library (Wolf et al., 2020). The entire workflow from data collection to model training is depicted in Fig. 1.

4. Results

4.1. Dataset characteristics

Figure 2 displays the class distributions within the two datasets we study. The class distributions are different, where Fin-SoMe has a strong positivity bias and only contains very few bearish posts. In the dataset we collected from Twitter, the most prevalent class is *neutral*. However, there are still more positive than negative tweets. These differences might also originate from labeling errors or divergent label definitions

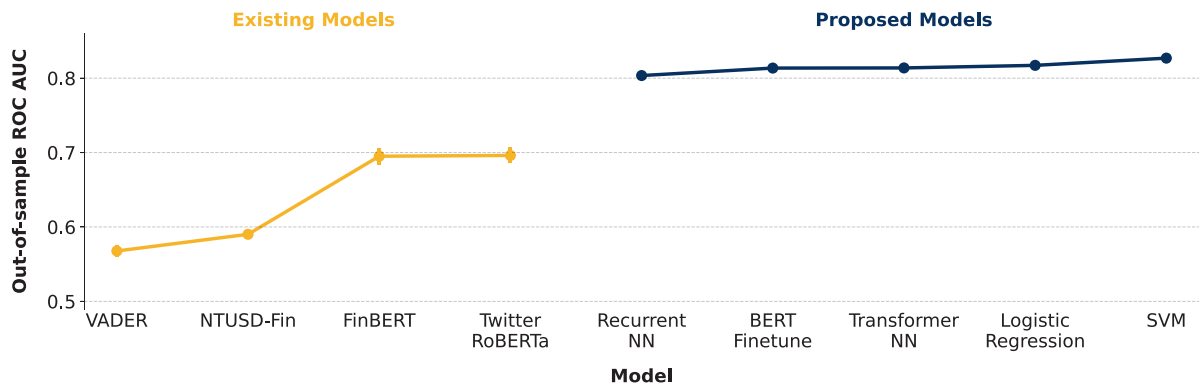

Fig. 3. Out-of-sample performance of existing models and proposed models on the collected dataset.

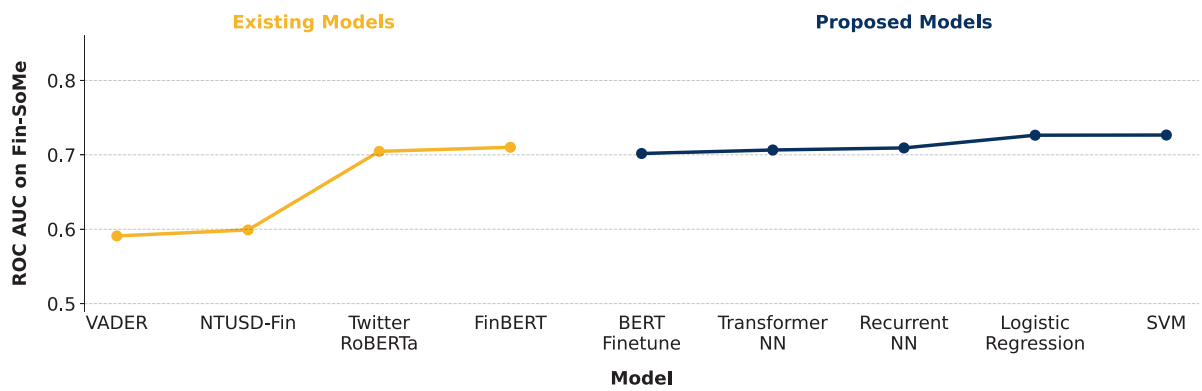

Fig. 4. Performance of models on the Fin-SoMe dataset.

in Fin-SoMe. For example, the message "$NXT.X December 28th is the key date. Dec 25–28 this is gonna be wild!" is labeled as bullish although the text sentiment is ambiguous as "wild" does not have a positive or negative connotation.

4.2. Model performance

We evaluate all models' ROC AUC on the collected dataset. Figure 3 presents the average and standard deviation (where applicable) across splits of all models' out-of-sample ROC AUC scores. The two dictionary-based models perform worst but beat random guessing. The finance-specific NTUSD-Fin lexicon (AUC = 0.59) beats VADER (AUC = 0.57), but not by a large margin. The two deep-learning-based models perform significantly better with AUC values of around 0.70. FinBERT and TwitterRoBERTa perform almost identically, although FinBERT has been trained on finance-specific data and TwitterRoBERTa has not. All of our proposed models outperform the existing ones on this dataset with AUC scores of above 0.80. Out of the proposed models, the recurrent network performs worst (AUC = 0.80). The fine-tuned BERT and transformer neural network perform better than the recurrent net (AUC = 0.81). Surprisingly, the simpler machine-learning-based logistic regression (AUC = 0.82) and support vector machine (AUC = 0.83) both perform even better than the neural networks. Their accuracy for the three-class classification task is around 64%.

Next, we evaluate the same models on an existing dataset, Fin-SoMe (Chen et al., 2020), and present their scores in Fig. 4. The performance of the existing models does not change much although VADER performs slightly better on this set of data (AUC = 0.59). The performance of our proposed models, on the other hand, degrades significantly as they now achieve AUC scores between 0.70 and 0.73. However, they still perform as well as or better than the best existing models. Our fine-tuned BERT model now performs slightly worse than the existing large language models, closely followed by the transformer-based and recurrent neural network. The logistic regression and support vector machine still outperform the neural networks suggesting that their tendency to overfit less is helping them generalize to unseen datasets.

The relative performance of the logistic regression and support vector machine is especially relevant when considering the computational cost that training and deploying these models entails. To illustrate this issue, we plot the inference time per sample in milliseconds in Fig. 5. Note that the y-axis is log-scaled due to the multiple orders of magnitude that lie between the inference times of the fastest and the slowest model. All experiments were conducted on a system with an AMD Ryzen 5 3600 CPU and 64 GB of RAM. The dictionary-based VADER and logistic regression model are the fastest at inference times below 0.1 ms/sample. Both neural networks we trained from scratch are significantly slower but still need less than 1 ms for each inference. The NTUSD-Fin lexicon and support vector machine perform similarly. Finally, all BERT-based deep learning models are up to 1000 times slower than the fastest model at around 100 ms/sample. Their large BERT architecture for encoding texts requires substantial compute time. While the lack of a GPU for these experiments slows down the deep-learning-based models significantly, the estimated speed-up obtained by running these models on specialized hardware is around 4–5x (Buber & Banu, 2018) which still leaves them behind all other models in terms of speed.

4.3. Model diagnostics

Considering the trade-off between model performance and training/inference time, we find the logistic regression model to strike the best balance between the two. While the SVM model scores a slightly higher AUC, it is around 50 times slower in both training and inference and not as interpretable. Consequently, in this section, we scrutinize the proposed logistic regression model and compare its behavior to the existing models.

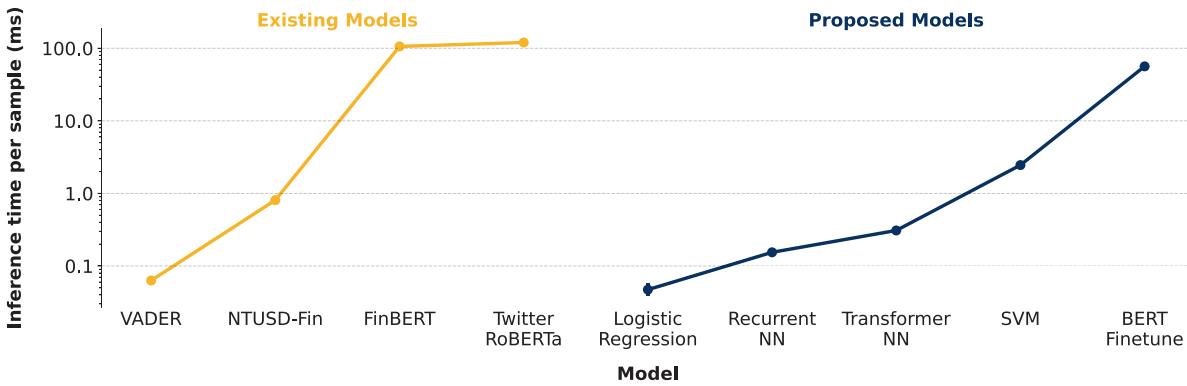

Fig. 5. Inference time per sample (ms, log-scaled) for existing and proposed models.

Table 3
The top 15 tokens with largest model coefficients for each of the three classes.

Class	Tokens associated with largest coefficients
Bullish	run, buy, rip, cal, call, 999c, bull, ulli, bul, llis, lish, ath, 🚀, up, buy
Neutral	tick, play, hart, name, hit, |, =, real, ser, ▦, 9:9, sur, er?, or, chat,
Bearish	fall, eari, dump, dum, rish, lowe, dow, shor, low, red, hort, 999p, down, los, put,

Table 4
The models predictions on example tweets organized by common topics. Numbers are predicted probabilities for the *correct class*. ***x.xx**** indicates that the model predicted the correct class.

Example tweet	y	$\mathbb{P}(\hat{y} = y)$				
		VADER	NTUSD	FinBERT	RoBERTa	pyFin
Stock Ownership						
adding to my $AAPL position	POS	0.00	***0.52****	***0.66****	0.21	***0.91****
I'm long $AAPL	POS	0.00	***1.00****	0.05	0.44	***0.91****
getting rid of my $AAPL position	NEG	0.00	***0.56****	0.05	0.16	0.36
Just shorted $AAPL	NEG	0.00	***0.65****	0.04	0.24	***0.97****
Options Trading						
Going all in $TSLA 4/20 $69 calls today before close	POS	0.00	***0.83****	0.10	0.08	***0.83****
Sold a 58P on $INTC two weeks out	POS	0.00	***0.78****	0.16	0.06	0.03
Going all in $TSLA 4/20 $69 puts today before close	NEG	0.00	0.46	0.03	0.01	***0.93****
Sold a 58C on $INTC two weeks out	NEG	0.00	0.22	0.02	0.04	0.43
Business Acumen						
$TSLA factory can start production sooner than expected	POS	0.00	***0.75****	0.11	***0.81****	***0.75****
$F beats EPS estimate, expected 1.34 reported 1.89	POS	0.00	***0.93****	0.07	***0.53****	***0.63****
$NFLX missed earnings estimates	NEG	0.42	0.34	***0.93****	***0.61****	0.28
$OXY to lay off 42% of staff leaked memo reveals	NEG	0.20	0.10	***0.96****	***0.49****	0.29
Neutral						
$OXY stocks trading at $123	NEU	***1.00****	0.00	***0.80****	***0.87****	***0.77****
$MMM Q3 numbers will decide the future of this stock	NEU	***1.00****	0.00	***0.93****	***0.86****	***0.54****
Come join our chatroom for exclusive stock tips!! $CASH	NEU	***0.62****	0.00	***0.92****	0.39	***0.87****
Kathryn Janeway to take over as new $SBUX CEO	NEU	***1.00****	0.00	***0.93****	***0.93****	0.38

For each of the three sentiment classes, Table 3 lists the tokens with the largest coefficients. These tokens, if present in a document, have the largest effect on predicting in favor of each of the classes. We observe that the model has learned domain-specific vocabulary, where words like "buy", "call", "run" indicate positive sentiment, and words like "dump", "lower", or "short" indicate negative sentiment. Moreover, it has learned that numeric patterns like "123C" or "123P" (call option or put option with a strike price of $123) express positive or negative sentiment respectively. For the positive class, the model has even picked up that a rocket emoji or abbreviations like "ATH" (all-time high) are indicative of positive market sentiment. Additionally, we see that subword tokenization helps with picking up different spellings of the same concept. For example, the word "bullish" can be spelled "buuullish" or "bullllish" by users who want to emphasize their opinion, which subword-based models can exploit for making more accurate predictions.

To put the predictive power of each model into perspective, we demonstrate their predictions on a set of example tweets. We categorize them by topics that frequently emerged in the collected data. While these examples do not perfectly represent the raw data, it demonstrates how each model copes with texts of varying difficulty from specific subtopics of online investing discussions. Table 4 shows an example for the four categories "stock ownership" (simple texts that do not mention complex financial instruments), "options trading" (texts on more complex financial instruments using implicit negation), "business acumen" (general business news), and "neutral" tweets which do not carry any sentiment.

We see that VADER, which relies on generic positive or negative words, is not able to pick up any domain-specific sentiment at all. Mostly, it predicts a neutral sentiment for the lack of any generic sentiment-laden words like "nice", "great", or "happy". NTUSD-Fin does not exhibit this neutrality bias, especially when a document contains clear domain-specific keywords like "long" or "short". However, it does not classify documents correctly which convey sentiment through abbreviations ("58C") or implied sentiment ("lay off staff") as it relies on

single sentiment-laden words. Similar to VADER, FinBERT also exhibits a bias towards predicting neutral, but performs better when documents resemble news headlines, potentially because it has been trained on similar data. TwitterRoBERTa performs similarly, although it has only been trained on generic social media posts. Finally, our proposed logistic regression model ("pyFin") outperforms the other models in most categories. Its weak points are tweets resembling news headlines and the implicit negation present in some of the options trading tweets. For example, while buying put options expresses a negative market sentiment, selling put options equates to bullish sentiment. Our model seems to base its prediction on the tokens "sell" and "99P", both of which are negative and predicts a negative sentiment with high confidence. It has not learned complex dependencies between words, which, for a linear bag-of-words model, is mathematically impossible. Therefore, this shortcoming has to be accepted when working with such models.

5. Discussion

Motivated by the lack of functioning model artifacts that provide accurate assessments of sentiment in finance-related social media posts, the goal of this study is to provide researchers with a tool to automate sentiment analysis in this specific domain. The results demonstrate that we were able to develop an artifact that achieves the study's objective of performing market sentiment analysis on tweets. The proposed logistic regression model outperforms all existing models from adjacent domains when applied to data from Twitter. Even when applied to StockTwits posts, we show that it outperforms BERT-based large language models that require up to 1000 times more compute capacity for inference alone. The findings confirm that for the task of domain-specific sentiment analysis small models can outperform more complex ones if trained on data from the domain that they will be applied to at inference time. Depending on the amount of labeled data that is available for this task, the results indicate that complex models are prone to overfitting which limits their ability to generalize. We observe this during training, where the accuracy on the training data is significantly higher than on the validation set, and when applying them to the Fin-SoMe dataset. Under these circumstances, the simpler logistic regression and SVM still performed best. This highlights the importance of simple machine learning models for learning tasks based on small datasets. While we demonstrate that dictionary-based models might be *too* simple to properly handle the text data we study, machine learning models strike a good balance between computational complexity and performance. This contradicts findings by Sohangir et al. (2018) who find that a convolutional neural network outperforms a logistic regression on the two-class classification task of predicting sentiment in StockTwits messages. However, they have access to six months worth of labeled data which helps alleviate overfitting issues in deep learning models and enables them to score an AUC score of 0.90. Therefore, further research is needed to determine which model class is best suitable for datasets of different sizes.

Apart from the sentiment model type, our results confirm findings by Ravi & Ravi (2015) who suggest researchers need to be careful when applying sentiment analysis models across domains. We show that none of the existing models generalize well to Twitter content and that the predictive performance of models trained on tweets degrades significantly when applied to StockTwits messages. This implies that Twitter and StockTwits should not be used as interchangeable data sources as the nature of posts published on both platforms varies. However, the StockTwits messages in Fin-SoMe are already multiple years old. As the economic situation and hence online discussion topics change over time, these differences might be partially attributed to the different sampling time frames.

An error analysis of our proposed logistic regression model suggests that it struggles with implicit, domain-specific negation when classifying word groups like "selling a put". This coincides with the sentiment analysis challenges laid out by Hussein (2018) who lists implicit and explicit negation as one of the toughest challenges. The model lacks the ability to capture dependencies between words within a document. Transformer-based neural networks promise to address this shortcoming and should theoretically be able to learn that the combination of the word "sell" and "put" conveys positive market sentiment. Our results demonstrate that they were not able to deliver on that promise on the small dataset we studied in this work due to overfitting issues.

Finally, our results suggest that extracting market sentiment from tweets is a harder task than sentiment analysis of news headlines. BERT-based models can classify news headlines with accuracies of up to 86% (F1 score: 0.84) (Araci, 2019) while our best model achieves an accuracy of around 64% (F1 score: 0.63). A potential reason for this is that tweets contain more ambiguous messages, sarcasm, or slang words than editorial content like news headlines.

5.1. Contributions to the literature

The findings established by this work add to the literature on sentiment analysis as well as research that is using social sentiment to study other phenomena. From a theoretical point of view, we show that the choice of sentiment model can dramatically affect the quality of sentiment assessments. In particular, dictionary-based models do not perform well on the domain-specific tweets we study. Even though they are fast and easy to use our results demonstrate that machine-learning-based models can deliver better predictions at similar resource requirements. This implies that scholars who use automated sentiment assessments should consciously choose a model that has been shown to perform well on the type of data they are working with as generic off-the-shelf models like VADER can perform poorly on domain-specific texts. Considering that research often utilizes sentiment analysis for a specific domain or industry, our findings raise the question of whether generic sentiment analysis models should be replaced by domain-specific models in other applications too. Kumar, Kar, & Ilavarasan (2021) show that the majority of the literature using sentiment analysis studies domains like the hotel, restaurant, business, sales, or tourism industry. Like finance and investing, all these domains use a plethora of specific jargon and terms that a generic sentiment model cannot capture, thus yielding sub-par sentiment predictions.

5.2. Practical implications

This work demonstrates the process to design a custom sentiment model artifact for sentiment analysis of finance-related tweets (RQ1). Our findings can guide future research in the field of sentiment analysis, e.g. when designing custom models for other domains. For the domain of finance and retail investing, we show that our model artifact outperforms previously existing models at a fraction of the computational cost (RQ2 & RQ3). The comparison of model performance on a dataset from Twitter and StockTwits demonstrates that the source of the data the models are applied to can significantly impact their performance as well, thus researchers should only rely on a sentiment model's prediction when it is applied to data similar to the one it was trained on (RQ4). Using the proposed sentiment model, future research could reassess results based on sentiment scores from generic models to scrutinize how more accurate sentiment assessments affect the results in downstream analyses like stock volatility and return prediction (Ren et al., 2018; Wilksch & Abramova, 2022) or use sentiment as an additional independent variable in studying phenomena like anomalous stock price movements (Al-Sulaiman, 2022). Besides using sentiment as an explanatory variable in predictive modeling, it can also be useful to researchers studying the clustering of different stocks (Gonzales & Hargreaves, 2022). Considering that different kinds of investments attract more or less conservative investors the investor sentiment could be a helpful indicator to study the similarity between stocks.

5.3. Limitations

Limitations of this work include its focus on English tweets about large US corporations. Our results need not generalize to foreign equity markets, cryptocurrency markets, or small-cap stocks and only capture sentiment of people who share their opinions on Twitter. Moreover, the data we collect is limited to a year's worth of tweets, hence model performance might suffer if applied to data from other platforms or future discussions. Therefore, future research could examine how the predictive performance of a sentiment model decays over time to gauge the need for constant retraining of such models. Additionally, our results regarding the performance of neural networks can be re-assessed on a larger dataset or a dataset labeled by multiple labelers which would allow studying the inter-rater-reliability of this task. Finally, an emergent issue our work is subject to is fake news on social media platforms. While there have been many attempts to automatically detect fake news (Ansar & Goswami, 2021), any indicator that is based on social media content can be manipulated by bad actors posting on large platforms. On the other hand, findings by Aswani, Kar, & Ilavarasan (2019) suggest that a text's emotion and polarity can indicate how authentic it is. Therefore, future research could potentially even use sentiment analysis to further the field of fake news detection.

To facilitate future research on financial market sentiment in social media posts, we publish our logistic regression model as an easy-to-use python library. The library is available under an MIT license on the python package index at https://pypi.org/project/pyfin-sentiment/. Using the package installer for python (pip), it can be installed with one command: `pip install pyfin-sentiment`. The library is documented at https://pyfin-sentiment.readthedocs.io/en/latest.

6. Conclusion

In this work, we address the issue of domain-specific sentiment analysis of finance-related social media posts. We show that existing models trained on finance-related texts or generic social media posts do not perform well when applied to documents from this specific subfield. By collecting and annotating a dataset of 10,000 tweets, we design, implement, and deploy a machine learning model that is capable of performing this task effectively and efficiently. Despite its simple architecture, it outperforms BERT-based large language models trained on adjacent tasks, recurrent and transformer-based neural networks we train from scratch, and even the BERT network we fine-tune on our dataset. We highlight each model's strengths and shortcomings and publish our model artifact as a python library to foster future research.

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Appendix A

The search query used for obtaining data from Twitter's API.

```
($TSLA OR $TWTR OR $AAPL OR $NFLX OR $FB OR
$AMZN OR $GM OR $AMD OR $NVDA OR $MSFT OR $DIS
OR $GOOGL OR $F OR $GOOG OR $PYPL OR $CAT OR $T
OR $CVX OR $BAC OR $AAL OR $BA OR $PFE OR $INTC
OR $JPM OR $OXY OR $ES OR $WMT OR $UAL OR $DAL
OR $C OR $KO OR $XOM OR $COST OR $CCL OR $MRNA
OR $MU OR $GS OR $WFC OR $QCOM OR $JNJ OR $MS
OR $CRM OR $SBUX OR $VZ OR $ABBV OR $V OR $MMM
OR $WBD OR $NCLH OR $PG) lang:en -is:retweet
-is:nullcast -has:images -has:videos
```

Appendix B

Table 5
The codebook that guides the data labeling.

Bullish	Bearish
• bought stock, holding stock, *not* selling stock, want to buy stock, regret selling	• selling a stock, even when locking in profits
• buying calls, selling puts, being long	• buying puts, selling calls, being short
• stock is a bargain, undervalued, oversold, reaching all-time high, is in an up trend	• not buying or selling for negative expectations
• positive earnings release, growing revenue, profits, or customer base, *not* absolute numbers without judgement or direction	• stock is overvalued, overbought, reaching a new low, is in a down trend
• price target raised	• negative news like lawsuits, layoffs, or bad press
• praising or using stock as a positive example, asking positive rhetorical question	• lowered price target
• business acquisitions & expansions, product launches	• insulting, mocking, or using stock as a negative example
• being excited, liking, using, or buying a company's products & services	• asking rhetorical questions suggesting negative sentiment
	• disliking, banning, avoiding, or stop using a company's products & services
Neutral, uncertain, no sentiment	
• not investing for uncertain expectations	• any absolute numbers without clear directional interpretation
• list pro *and* con arguments for investment	• neutral information or news headlines
• list positive *and* negative opinions or facts in same tweet, also when referring to two or more different stocks	• stating non-opinionated facts that are not inherently positive or negative
• asking for guidance because of uncertainty	• changes in volatility
• looking forward to, keeping an eye on, or generally being interested in future events	• spam, ads, or not related to topic of investing
• buying/selling straddles, strangles, iron condors	• seeking other's opinions, asking a question
• sentiment is present, but it is unclear whether it is positive or negative	• stating an opinion which does not contain a positive or negative sentiment
• *not:* changing one's mind, the more recent opinion is used as the label	• post is about cryptocurrency

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jjimei.2023.100171.

References

Aggarwal, D. (2019). Defining and measuring market sentiments: A review of the literature. *Qualitative Research in Financial Markets, 14*(2), 270–288.

Ahuja, R., Rastogi, H., Choudhuri, A., & Garg, B. (2015). Stock market forecast using sentiment analysis. In *2015 2nd international conference on computing for sustainable global development (INDIACom)* (pp. 1008–1010). IEEE.

Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25rd ACM SIGKDD international conference on knowledge discovery and data mining*.

Al-Shabi, M. (2020). Evaluating the performance of the most important lexicons used to sentiment analysis and opinions mining. *International Journal of Computer Science and Network Security, 20*(1), 1.

Al-Sulaiman, T. (2022). Predicting reactions to anomalies in stock movements using a feed-forward deep learning network. *International Journal of Information Management Data Insights, 2*(1), 100071.

AlBadani, B., Shi, R., & Dong, J. (2022). A novel machine learning approach for sentiment analysis on Twitter incorporating the universal language model fine-tuning and SVM. *Applied System Innovation, 5*(1), 13.

Ansar, W., & Goswami, S. (2021). Combating the menace: A survey on characterization and detection of fake news from a data science perspective. *International Journal of Information Management Data Insights, 1*(2), 100052.

Antweiler, W., & Frank, M. Z. (2004). Is all that talk just noise? The information content of internet stock message boards. *The Journal of Finance, 59*(3), 1259–1294.

Araci, D. (2019). FinBERT: Financial sentiment analysis with pre-trained language models. arXiv preprint arXiv:1908.10063

Aswani, R., Kar, A. K., & Ilavarasan, P. V. (2019). Experience: Managing misinformation in social media–insights for policymakers from Twitter analytics. *Journal of Data and Information Quality (JDIQ), 12*(1), 1–18.

Audrino, F., Sigrist, F., & Ballinari, D. (2020). The impact of sentiment and attention measures on stock market volatility. *International Journal of Forecasting, 36*(2), 334–357.

Baccianella, S., Esuli, A., & Sebastiani, F. (2010). Sentiwordnet 3.0: An enhanced lexical resource for sentiment analysis and opinion mining. In *Proceedings of the seventh international conference on language resources and evaluation (LREC'10)*.

Barbieri, F., Camacho-Collados, J., Neves, L., & Espinosa-Anke, L. (2020). TweetEval: Unified benchmark and comparative evaluation for tweet classification. arXiv preprint arXiv:2010.12421

Bradley, M. M., & Lang, P. J. (1999). Affective norms for English words (ANEW): Instruction manual and affective ratings. *Technical report C-1*. The Center for Research in Psychophysiology ....

Buber, E., & Banu, D. (2018). Performance analysis and CPU vs. GPU comparison for deep learning. In *2018 6th international conference on control engineering & information technology (CEIT)* (pp. 1–6). IEEE.

Chen, C.-C., Huang, H.-H., & Chen, H.-H. (2018). Ntusd-fin: A market sentiment dictionary for financial social media data applications. In *Proceedings of the 1st financial narrative processing workshop (FNP 2018)*.

Chen, C.-C., Huang, H.-H., & Chen, H.-H. (2020). Issues and perspectives from 10,000 annotated financial social media data. In *Proceedings of the 12th language resources and evaluation conference* (pp. 6106–6110).

Chintalapudi, N., Battineni, G., Di Canio, M., Sagaro, G. G., & Amenta, F. (2021). Text mining with sentiment analysis on seafarers' medical documents. *International Journal of Information Management Data Insights, 1*(1), 100005.

Cho, K., Van Merrienboer, B., Bahdanau, D., & Bengio, Y. (2014). On the properties of neural machine translation: Encoder-decoder approaches. arXiv preprint arXiv:1409.1259

Cortis, K., Freitas, A., Daudert, T., Huerlimann, M., Zarrouk, M., Handschuh, S., et al., (2017). Semeval-2017 task 5: Fine-grained sentiment analysis on financial microblogs and news. In *Association for computational linguistics (ACL)* (pp. 519–535). Association for Computational Linguistics. 10.18653/v1/S17-2089.

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2018). Bert: Pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:1810.04805

Gonzales, R. M. D., & Hargreaves, C. A. (2022). How can we use artificial intelligence for stock recommendation and risk management? A proposed decision support system. *International Journal of Information Management Data Insights, 2*(2), 100130.

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT Press. http://www.deeplearningbook.org

Hirshleifer, D. (2015). Behavioral finance. *Annual Review of Financial Economics, 7*, 133–159.

Hussein, D. M. E.-D. M. (2018). A survey on sentiment analysis challenges. *Journal of King Saud University-Engineering Sciences, 30*(4), 330–338.

Hutto, C., & Gilbert, E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. In *Proceedings of the international AAAI conference on web and social media: vol. 8* (pp. 216–225).

Kudo, T. (2018). Subword regularization: Improving neural network translation models with multiple subword candidates. arXiv preprint arXiv:1804.10959

Kumar, S., Kar, A. K., & Ilavarasan, P. V. (2021). Applications of text mining in services management: A systematic literature review. *International Journal of Information Management Data Insights, 1*(1), 100008.

Liu, B., Hu, M., & Cheng, J. (2005). Opinion observer: Analyzing and comparing opinions on the web. In *Proceedings of the 14th international conference on world wide web* (pp. 342–351).

Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., & Chen, D. et al. (2019). Roberta: A robustly optimized bert pretraining approach. arXiv preprint arXiv:1907.11692

Loshchilov, I., & Hutter, F. (2017). Decoupled weight decay regularization. arXiv preprint arXiv:1711.05101

Loughran, T., & McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-ks. *The Journal of Finance, 66*(1), 35–65.

Luo, M., & Mu, X. (2022). Entity sentiment analysis in the news: A case study based on negative sentiment smoothing model (NSSM). *International Journal of Information Management Data Insights, 2*(1), 100060.

Mishev, K., Gjorgjevikj, A., Vodenska, I., Chitkushev, L. T., & Trajanov, D. (2020). Evaluation of sentiment analysis in finance: From lexicons to transformers. *IEEE Access, 8*, 131662–131682.

Mittal, A., & Goel, A. (2012). Stock prediction using Twitter sentiment analysis. Standford University, CS229 (2011 http://cs229.stanford.edu/proj2011/GoelMittal-StockMarketPredictionUsingTwitterSentimentAnalysis.pdf), 15, 2352.

Nielsen, F. Å. (2011). A new anew: Evaluation of a word list for sentiment analysis in microblogs. arXiv preprint arXiv:1103.2903

Obembe, D., Kolade, O., Obembe, F., Owoseni, A., & Mafimisebi, O. (2021). COVID-19 and the tourism industry: An early stage sentiment analysis of the impact of social media and stakeholder communication. *International Journal of Information Management Data Insights, 1*(2), 100040.

Oliveira, N., Cortez, P., & Areal, N. (2017). The impact of microblogging data for stock market prediction: Using Twitter to predict returns, volatility, trading volume and survey sentiment indices. *Expert Systems with Applications, 73*, 125–144.

Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., et al., (2019). Pytorch: An imperative style, high-performance deep learning library. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, & R. Garnett (Eds.), *Advances in neural information processing systems 32* (pp. 8024–8035). Curran Associates, Inc.. http://papers.neurips.cc/paper/9015-pytorch-an-imperative-style-high-performance-deep-learning-library.pdf

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al., (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12*, 2825–2830.

Pennebaker, J. W., Francis, M. E., & Booth, R. J. (2001). *Linguistic inquiry and word count: LIWC 2001*: vol. 71. Mahway: Lawrence Erlbaum Associates.

Ravi, K., & Ravi, V. (2015). A survey on opinion mining and sentiment analysis: Tasks, approaches and applications. *Knowledge-Based Systems, 89*, 14–46.

Ren, R., Wu, D. D., & Liu, T. (2018). Forecasting stock market movement direction using sentiment analysis and support vector machine. *IEEE Systems Journal, 13*(1), 760–770.

Renault, T. (2020). Sentiment analysis and machine learning in finance: Acomparison of methods and models on one million messages. *Digital Finance, 2*(1), 1–13.

Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). Distilbert, a distilled version of bert: Smaller, faster, cheaper and lighter. arXiv abs/1910.01108

Sohangir, S., Wang, D., Pomeranets, A., & Khoshgoftaar, T. M. (2018). Big data: Deep learning for financial sentiment analysis. *Journal of Big Data, 5*(1), 1–25.

Stone, P. J., & Hunt, E. B. (1963). A computer approach to content analysis: Studies using the general inquirer system. In *Proceedings of the May 21–23, 1963, spring joint computer conference*. In *AFIPS '63 (Spring)* (pp. 241–256). New York, NY, USA: Association for Computing Machinery. 10.1145/1461551.1461583.

Tang, D., Wei, F., Yang, N., Zhou, M., Liu, T., & Qin, B. (2014). Learning sentiment-specific word embedding for Twitter sentiment classification. In *Proceedings of the 52nd annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 1555–1565). Association for Computational Linguistics. 10.3115/v1/P14-1146.

Thelwall, M., Buckley, K., Paltoglou, G., Cai, D., & Kappas, A. (2010). Sentiment strength detection in short informal text. *Journal of the American Society for Information Science and Technology, 61*(12), 2544–2558.

Wilksch, M. V., & Abramova, O. (2022). The predictive power of social media sentiment for short-term stock movements. In *Wirtschaftsinformatik 2022 proceedings: vol. 38*.

Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., et al., (2020). Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 conference on empirical methods in natural language processing: System demonstrations* (pp. 38–45).

Yao, F., & Wang, Y. (2020). Domain-specific sentiment analysis for tweets during hurricanes (DSSA-H): A domain-adversarial neural-network-based approach. *Computers, Environment and Urban Systems, 83*, 101522.